

Tab 1

11.Hospital Appointment Booking System

(Using Core Java & OOP Concepts)

Under the Guidance of

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Abstract

The Hospital Appointment Booking System is a Java-based project developed using Object-Oriented Programming (OOP) concepts. The main purpose of this system is to efficiently manage the scheduling of appointments between doctors and patients. In traditional systems, managing appointments manually can lead to confusion, double bookings, and inefficiency.

This project provides a structured way to store doctor and patient details, check time slot availability, and manage appointments. It uses multiple classes such as Doctor, Patient, Appointment, and AppointmentManager to demonstrate real-world application of OOP concepts like class interaction and object composition.

The system helps in reducing manual work, improving scheduling accuracy, and enhancing overall efficiency in hospital management.

Introduction

The Hospital Appointment Booking System is designed to simplify the process of booking and managing medical appointments. The system allows users to store doctor and patient information, book appointments, cancel them, and display scheduled appointments.

Objectives

- Manage doctor and patient records
- Book and cancel appointments
- Avoid double booking of time slots
- Apply OOP concepts like classes and objects
- Reduce manual errors in scheduling

Scope

This project is suitable for small hospitals or clinics. It can be extended in future by:

- Adding database integration
 - Creating a graphical user interface (GUI)
 - Adding online booking features
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System Requirements

Hardware:

- Computer with minimum 4GB RAM

Software:

- Java JDK
 - IDE (VS Code / Eclipse)
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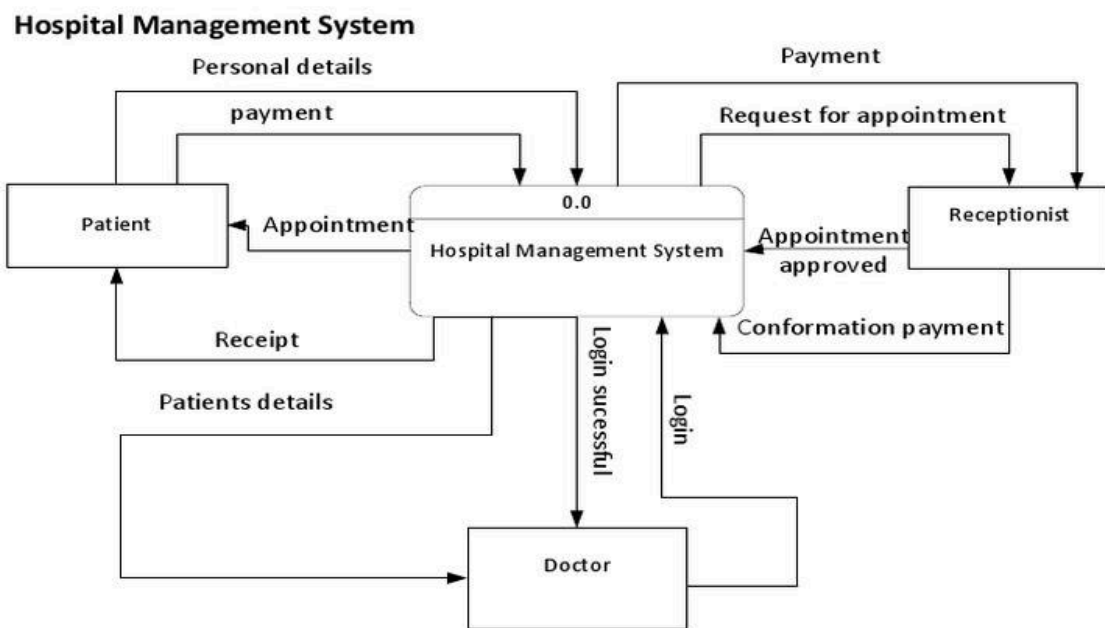
Methodology

The system works in the following steps:

1. Create doctor and patient records
2. Store data in arrays
3. Select doctor and time slot
4. Check availability of time slot
5. Book appointment if slot is available
6. Cancel appointment if required
7. Display all appointments

This structured approach ensures efficient scheduling and avoids conflicts.

Flowchart



DFD Level 0

- Book Appointment
 - Cancel Appointment
 - View Appointments → End
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Class Diagram

Classes used in the project:

Doctor Class

- Variables: ID, Name, Specialization
- Methods: Get details

Patient Class

- Variables: ID, Name, Age
- Methods: Get details

Appointment Class

- Variables: Doctor, Patient, Time Slot, Status
- Methods: Book, Cancel

AppointmentManager Class

- Methods:
 - Check availability
 - Book appointment
 - Cancel appointment
 - Display appointments

Implementation

The project is implemented using multiple interacting classes:

- **Doctor & Patient classes** store basic information
- **Appointment class** links doctor and patient
- **AppointmentManager class** handles all operations

Logic

- System checks if slot is available
- If yes → appointment booked
- If no → error message shown
- Appointments can be cancelled using ID or time slot

Sample Input/Output

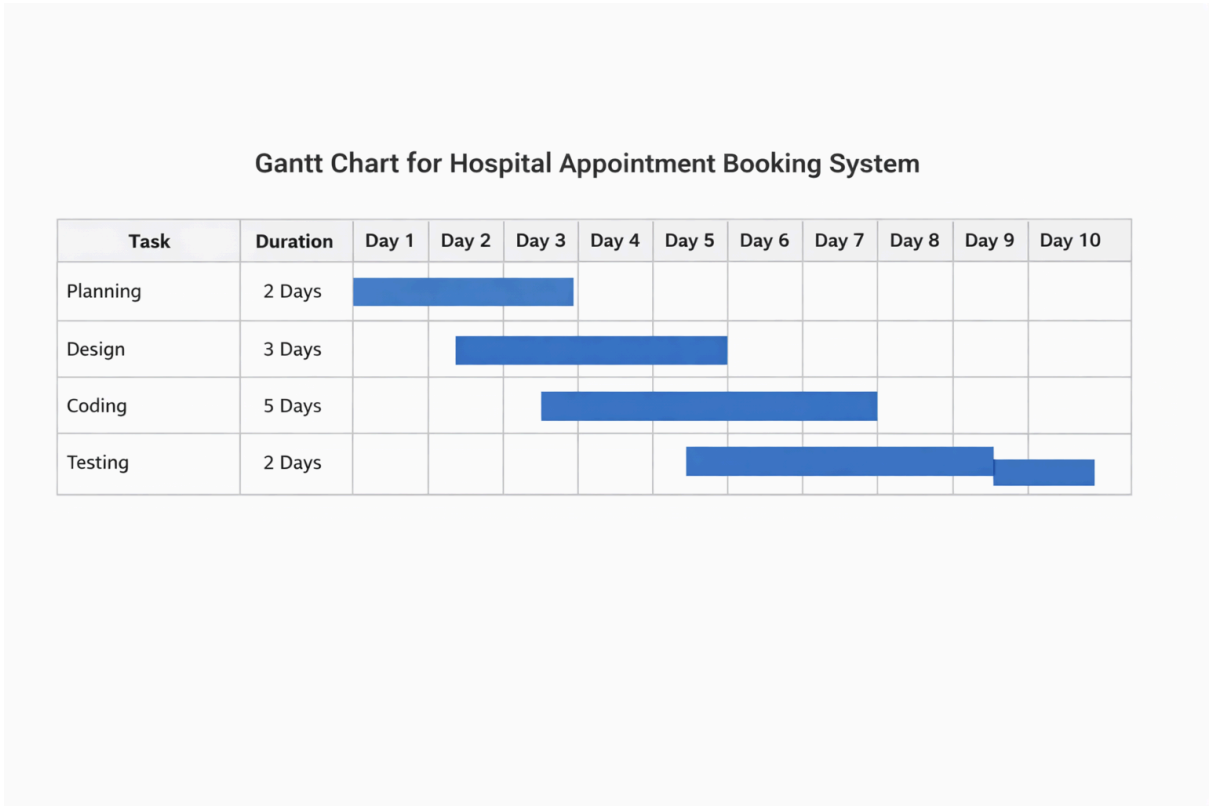
Example:

- Appointment booked successfully
- Slot already booked
- Appointment cancelled

Future Enhancements

- Add database (MySQL)
- Develop GUI (Java Swing)
- Add login system
- Enable online booking

Gantt Chart



Conclusion

The project successfully demonstrates the use of Object-Oriented Programming concepts in a real-world application. It provides an efficient way to manage hospital appointments and reduces manual errors.

References

Books:

[1] H. Schildt, *Java: The Complete Reference*, McGraw-Hill, 2018.

Website:

[2] Oracle, "Java Documentation," [Online]. Available: <https://docs.oracle.com>